Anusha Kumaresan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.41
Bangalore International Academy, Jayanagar	Class XII (CBSE)	2023	96.20%
Sri Kumaran Children's Academy, Doddakalasandra	Class X (CBSE)	2021	96.60%

EXPERIENCE

Databricks -- *Software Engineering Intern*

- Worked on the **Ingestion team**, modernizing **CDC setup validation** using **Scala, Java, JDBC & SQL Server metadata**; re-architected it onto **Scala/Spark QBC**, eliminating **~7-9 min gateway cold-start wait** and enabling **live-cluster execution**.
- **Productionized validation workflow** and integrated **AI-agent tooling** for on-demand validation diagnostics and remediation guidance.
- Proactively identified and **resolved 5-6 existing production bugs** and delivered **3 additional improvements** beyond the core project scope.

SCHOLASTIC ACHIEVEMENTS

- **Micron Global Women Mentorship Program**: Among 48 students selected nationwide; completed mentorship and industry workshops.
- **5th Place, Inter-IIT Tech Meet 14.0**: Represented IIT Hyderabad in Adobe's AI Product Development Challenge
- **Semifinalist in Google Girl Hackathon**: Selected from 50,000+ students; advanced through multiple coding rounds.
- **KVPY SA 2021**: Qualified in Stream SA, securing AIR 1345; **NTSE 2021**: Ranked 20th at the state level in Stage-1 of the exam
- **KCET 2023**: Secured rank 119; **VITEEE 2023**: Secured national rank 310

PROJECTS

Fault Diagnosis in Distributed Training | Research under Dr. Praveen Tammana

- Developed a lightweight framework to **detect and localize fail-slow straggler GPUs** using CPU, GPU, network & training-level metrics.
- Engineered an instrumentation pipeline using **system profiling tools, PyTorch Profiler, and NVML** to capture per-iteration metrics and construct provenance graphs in real-time, modeling CPU, CUDA, and NCCL dependencies for causal analysis and root-cause diagnosis.

Concurrent Synchronization & Data Structures

- Built **lock-free queues** and **multi-reader single-writer registers** for thread-safe concurrent access.
- Implemented **TAS, TTAS, CLH, and MCS spinlocks**, comparing concurrent list performance with static & dynamic thread allocation.

Finance DSL Compiler (Finex) | Dr. Ramakrishna Upadrasta

- Designed and implemented a **statically typed financial DSL** and **compiler** using **C++, Flex, and Bison** under the Compilers course.

- Built a multi-phase compiler with LALR(1) parsing, AST construction, semantic analysis, type checking, and scoped symbol resolution.

RISC-V Simulator | Dr. Rajesh Kedia

- Developed a **RISC-V ISA simulator in C++,** as part of the Computer Architecture course, that parses assembly, encodes and executes instructions, and visualizes **register and memory states** with configurable cache simulation.
- Implemented a terminal-based interface with single-step execution, breakpoints, syntax validation, and error handling.

Secure Client-Server File Transfer System | Dr. Saurabh Kumar

- Built a **secure multi-client TCP file-transfer system** with authenticated file operations and a custom TLV application-layer protocol.
- Secured using SHA-256 authentication, Diffie-Hellman key exchange, AES-128 encryption, and MAC-based integrity verification.

Virtual Memory Management in gemOS | Dr. Saurabh Kumar, Dr. Satya Peri

- Designed and implemented **memory management features in gemOS for x86** as part of the Operating Systems course.
- Built VMA handling, lazy allocation, 4-level page-table fault handling, copy-on-write via mmap, munmap, mprotect, cfork system calls.

Intent-Based Networking (IBN) | Dr. Praveen Tammana, Dr. Kotaro Kataoka

- Designed an IBN system translating **natural-language intents** into network policies via **RASA NLU** for host extraction & classifying intent.
- Integrated an **SDN controller** for Dijkstra-based path computation and dynamic rule installation via **P4Runtime**.

POSITIONS OF RESPONSIBILITY

- **Dev Core at Lambda**, The Software development club of IITH Aug

- **Core at Kludge**, The Information Security and Networking Club of IITH

- **Core at Epoch**, The AI-ML and Data Science Club of IITH

- Teaching assistant for the course **Intro. to Artificial Intelligence**

SKILLS

Languages: C, C++, Python, HTML, CSS, JavaScript, SQL, Scala, Dart, PHP | **Technologies**: React.js, Node.js, Express.js, Flutter, MySQL, MongoDB, PyTorch, SocketIO, Google App Script | **Tools and Platforms**: Linux, Git, GitHub, Bash, VS Code, LaTeX, Markdown

RELEVANT COURSES

Computer Science: Data Structures, Algorithms, Software Engineering, Software Development Fundamentals, Operating Systems, Intro. to AI, Foundations of Machine Learning, Deep Learning, Parallel & Concurrent Programming, Database Management Systems, Computer Networks, Network Security, Computer Architecture, Compilers, Theory of Computation, Introduction to Programming, Discrete Mathematics

Mathematics: Linear Algebra, Probability and Random Variables, Statistics, Calculus, Linear Optimization, Transform Techniques

EXTRACURRICULAR

- Badminton Player: Represented IITH at Inter IIT Sports Meet 2023, 2024; Winner of multiple badminton tournaments & member of NSO

- Assistant Manager at E-Cell, IIT Hyderabad, Corporate Relations and Finance Team